ST5202 — Applied Regression Analysis Lecture Notes 3. Linear Regression Models for Analysis of Variance

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Outline

§1. The ANalysis Of VAriance (ANOVA) problems

§2. Traditional approch to ANOVA

§3. Regression approach to ANOVA

§1.1 Problems with one factor

- In practical fields, often several different treatments to a common problem need to be compared in order to find out the differences among the treatments, e.g., different therapies to treat a medical problem, different materials to manufacture a consumer product, etc.
- The different treatments are the categorical values of a factor.
- A factor is a variable whose values have no numerical meanings. The values of a factor are referred to as the levels of the factor.
- The study has data with the one-way layout below and is referred to as an one-way ANOVA problem which has only one factor.

Treatment 1	$\cdots$	Treatment a
y_{11}	$\cdots$	y_{a1}
$\cdots$	$\cdots$	$\cdots$
y_{1n_1}	$\cdots$	y_{an_a}
μ_1	$\cdots$	μ_a

- The purpose of one-way ANOVA is to investigate the effects of the treatments on the response variable.

§1.2. Problems with two factors

- In certain practical problems, the effects of two factors need to be investigated simultaneously, e.g., the effects of selling price and type of promotion campaign on sales of a certain product, the effects of carbon content and temperature on the strength of steel, etc.
- The study has data with a two-way layout below and is referred to a two-way ANOVA problem.

Levels of Factor 1	Levels of Factor 2		
	1	$\cdots$	b
1	y_{111} $\cdots$ $y_{11n_{11}}$	$\cdots$ $\cdots$ $\cdots$	y_{1b1} $\cdots$ $y_{1b2n_{1b}}$
	μ_{11}	$\cdots$	μ_{1b}
	$\vdots$	$\vdots$	$\vdots$
a	y_{a11} $\cdots$ $y_{a1n_{a1}}$	$\cdots$ $\cdots$ $\cdots$	y_{ab1} $\cdots$ $y_{abn_{ab}}$
	μ_{a1}	$\cdots$	μ_{ab}

- The purpose of two-way ANOVA is to investigate the effects of each factor as well as their interactions.
- In general, more than two factors can be considered. The problem will be referred to as a multi-way ANOVA problem.

§1.3. Measure of effects: contrasts

- A factor has effects on the response if the expected response changes when the levels of the factor changes.
- The effects of a factor are the differences (in a generic sense) of the expected responses at its different levels.
- A specific effect is measured by a contrast. A contrast is defined as
$$\sum_{k=1}^{a} c_k \mu_k \quad \text{with} \quad \sum_{k=1}^{a} c_k = 0, \ \text{i.e.,} \ \ \boldsymbol{c}^\top \mathbf{1} = 0,$$
where μ_k's are level means and $\boldsymbol{c} = (c_1, \ldots, c_k)$ is called a contrast vector (or simply called contrast for convenience).
- A contrast multiplied by a constant still measures the same effect and is considered as the same contrast.
- Examples of contrasts:
$$\mu_i - \mu_j; \ (\mu_1 + \mu_2)/2 - (\mu_3 + \mu_4 + \mu_6)/3.$$
- That there is no effect is equivalent to that all the contrasts are zero, that is,
$$\mu_1 = \mu_2 = \cdots = \mu_a \Leftrightarrow \sum_{i=1}^{a} c_k \mu_k = 0 \ \text{ for all } \boldsymbol{c}.$$

§1.4. Independent contrasts

- For a factor of a levels, all the contrast vectors form a subspace $\mathcal{R}^{a-1}$ of the Euclidean space R^a which is perpendicular to the vector $\mathbf{1} = (1, \cdots, 1)^\top$. Any $a-1$ linearly independent contrast vectors form a basis for $\mathcal{R}^{a-1}$. Any particular contrast can be expressed as a linear combination of the basis contrasts.
- A specific basis consists of

$$\boldsymbol{c}_1 = (-1, 1, 0, \ldots, 0)^\top, \ldots, \boldsymbol{c}_{a-1} = (-1, 0, \ldots, 0, 1)^\top,$$

  corresponding to

$$\mu_2 - \mu_1, \ldots, \mu_a - \mu_1.$$

- If a factor has only two levels, there is only one independent contrast, say, $\mu_1 - \mu_2$.

§1.5. Interaction effect: special cases

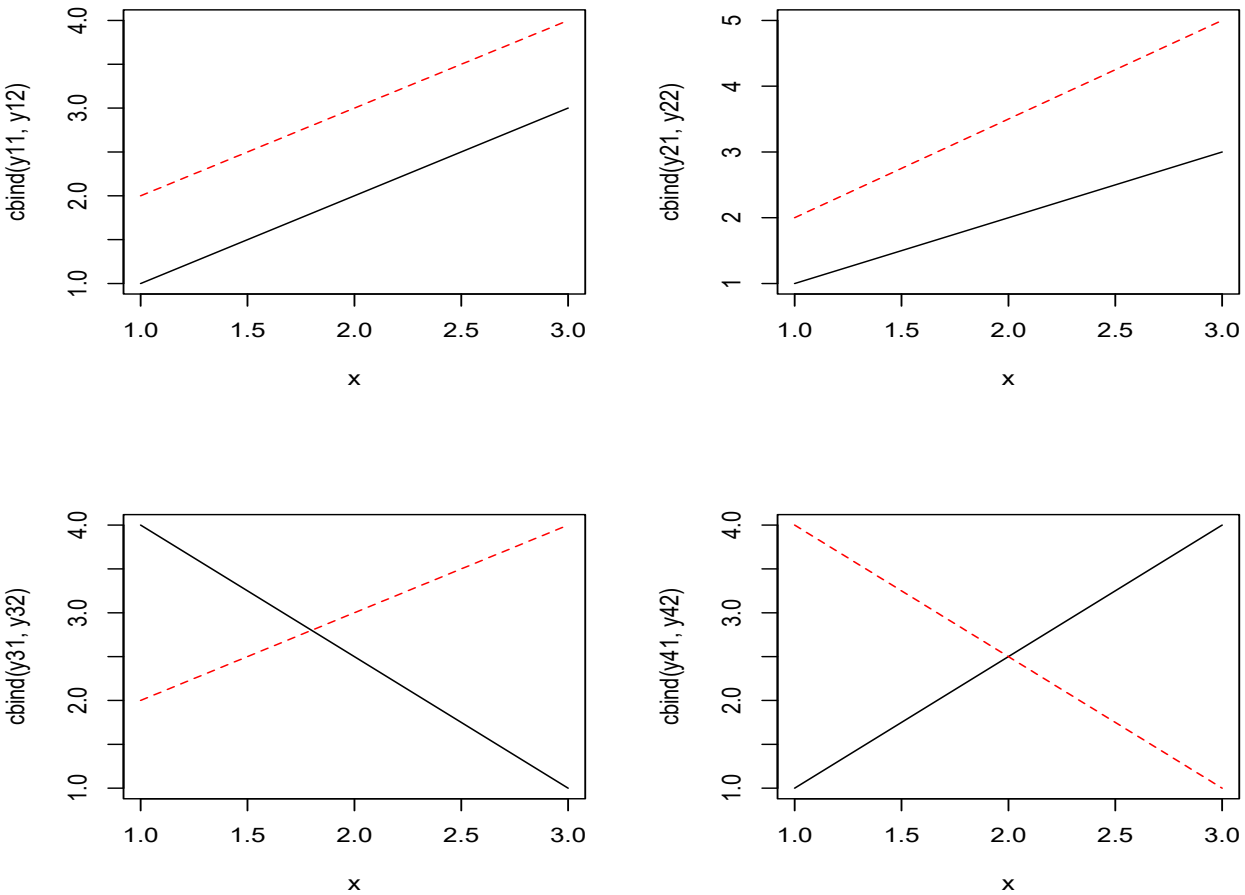
- If any effect of factor A is different when factor B is fixed at different levels, or, equivalently, if any effect of B is different when factor A is fixed at different levels, it is said that the two factors have interaction effect.
- If factor A and B have two levels each. The effects of A at the two levels of B are:

$$\mu_{21} - \mu_{11}, \quad \mu_{22} - \mu_{12}.$$

  If these two differences are not the same, there is interaction between A and B.
- The interaction effect is measured by

$$(\mu_{22} - \mu_{12}) - (\mu_{21} - \mu_{11}) = \mu_{22} - \mu_{12} - \mu_{21} + \mu_{11}.$$

§1.6. An illustration of interaction effects



§1.7. Interaction effect: general cases

- In general, an effect of factor A at a fixed level of B, say level i, is measured by a main-effect contrast $\sum_{k=1}^{a} c_k \mu_{ki}$. If there is at least one pair (i, j) and at least one contrast $\boldsymbol{c}$, such that
$$\sum_{k=1}^{a} c_k \mu_{ki} \neq \sum_{k=1}^{a} c_k \mu_{kj},$$
then A and B have interaction effect.
- An interaction effect is measured by a contrast of main-effect contrasts $\sum_{j=1}^{b} d_j \xi_j$, where $\xi_j = \sum_{k=1}^{a} c_k \mu_{kj}$ is a main-effect contrast of A at the jth level of B. It is called an interaction contrast.
- The interaction contrast is thus given by
$$\sum_{j=1}^{b} d_j \left[\sum_{k=1}^{a} c_k \mu_{kj} \right] = \sum_{j=1}^{b} \sum_{k=1}^{a} c_k d_j \mu_{kj}.$$
where $\sum_{k=1}^{a} c_k = 0, \quad \sum_{j=1}^{b} d_j = 0$.
- The vector of the coefficients in the double sum is called a interaction contrast vector. All the interaction contrast vectors form the interaction contrast space which is a subspace of R^{ab}.

§1.8. Independent interaction contrasts

- An interaction contrast vector has the structure:
$$(c_1 d_1, \ldots, c_1 d_b, \ldots, c_a d_1 \ldots c_a d_b),$$
with constraints:
$$\sum_{i=1}^{a} c_i d_j = 0, j = 1, \ldots, b; \ \sum_{j=1}^{b} c_i d_j = 0, i = 1, \ldots, a; \ \sum_{i=1}^{a} \sum_{j=1}^{b} c_i d_j = 0.$$
- Among the first b constraints, only $b-1$ are independent, among the second a constraints, only $a-1$ are independent, because of the last constraint. Altogether there are $a+b-1$ independent constraints. Thus the number of independent interaction contrasts is
$ab - a - b + 1 = (a-1)(b-1)$.
- Consider the $a-1$ independent contrasts of A at the fixed level j of B:
$$\mu_{ij} - \mu_{1j}, \ \ i = 2, \ldots, a.$$
Then, for each i, consider the $b-1$ independent contrasts across the levels of B:
$$(\mu_{ij} - \mu_{1j}) - (\mu_{i1} - \mu_{11}), \ \ j = 2, \ldots, b.$$
We then have $(a-1)(b-1)$ independent interaction contrasts:
$$\xi_{ij} = (\mu_{ij} - \mu_{1j}) - (\mu_{i1} - \mu_{11}), i = 2, \ldots, a; j = 2, \ldots, b.$$
Any interaction contrast can be expressed as a linear combination of these ξ_{ij}'s.

§2.1. Traditional approach to one-way ANOVA

- The traditional approach to the question whether there are any treatment effects is through the decomposition of the variation in the response variable into components caused by different sources.
- The variation of the response variable is measured by the total sum of squares (SST). It can be decomposed into the sum of squares attributable to the factor A (SSA) and the sum of squares attributable to errors (SSE).

$$\mathrm{SST} = \mathrm{SSA} + \mathrm{SSE}.$$

where

$$\mathrm{SST} = \sum_{i=1}^{a}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2, \ \mathrm{SSA} = \sum_{i=1}^{a} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2,$$

$$\mathrm{SSE} = \sum_{i=1}^{a}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2, \quad \bar{y}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij},$$

$$\bar{y}_{\cdot\cdot} = \frac{1}{n}\sum_{i=1}^{a}\sum_{j=1}^{n_i} y_{ij}, \quad n = \sum_{i=1}^{a} n_i.$$

§2.2. One-way ANOVA table

- The one-way ANOVA table is given by

Sourse	d.f.	SS	MS	F-test
A	a-1	SSA	MSA	MSA/MSE
Error	n - a	SSE	MSE	–
Total	n-1	SST	–	–

- The significance of the effect of the factor is tested by the F-statistic,

$$F = \frac{MSA}{MSE}.$$

Under the hypothesis $\mu_1 = \cdots = \mu_a$, $F \sim F_{a-1,n-a}$.

- To test the null hypothesis $\sum_i c_i \mu_i = 0$,(i.e., treatment comparison), the test statistic is given by

$$T = \frac{\sum_{i=1}^{a} c_i \bar{y}_{i\cdot}}{\sqrt{\hat{\sigma}^2 \sum_{i=1}^{a} c_i^2 / n_i}},$$

where $\hat{\sigma}^2 = MSE$. Under the null hypothesis, $T \sim t_{n-a}$.

§2.3. Kenton Food Company (KFC) Example: data

The Kenton Food Company wished to test four different package designs for a new breakfast cereal. Twenty stores, with approximately equal sales volumes, were selected as the experiment units. Each store was randomly assigned one of the package designs, with each package design randomly assigned to 5 stores. Sales, in number of cases, were observed for the study period. The data are in the following table:

	Stores							
Design	1	2	3	4	5	Total	Mean	Size
i	y_{i1}	y_{i2}	y_{i3}	y_{i4}	y_{i5}	$Y_{i\cdot}$	$\bar{Y}_{i\cdot}$	n_i
1	11	17	16	14	15	73	14.6	5
2	12	10	15	19	11	67	13.4	5
3	23	20	18	17		78	19.5	4
4	27	33	22	26	28	136	27.2	5
						$Y_{\cdot\cdot} = 354$	$\bar{Y}_{\cdot\cdot} = 18.63$	19

§2.4. KFC example: ANOVA

- Computation of SS:

$$
\begin{aligned}
SST &= (11-18.63)^2+\cdots+(28-18.63)^2 = 746.42, \\
SSA &= 5(14.6-18.63)^2+5(13.4-18.63)^2 \\
&\quad +4(19.5-18.63)^2+5(27.2-18.63)^2 = 588.22 \\
SSE &= 746.42-588.22 = 158.20.
\end{aligned}
$$

- ANOVA table:

Source	df	SS	MS	F
Factor A	3	588.22	196.07	18.6
Error	15	158.20	10.55	
Total	18	746.42		

- F-test: p-value `pf(18.6,3,15,lower.tail=FALSE) = 2.577736e-05`.
  Conclusion: The package design has significant effect on the sales, i.e., the sales for the four design packages are significantly different.

§2.5. Traditinal approch to two-way ANOVA

- Two-way ANOVA investigates whether there is interaction between the two factors,and whether the main effect is significant.
- The main-effect of a factor is the differences of the response averages at its different levels. For example, in case of both factors having only two levels, the main-effect of factor A is

$$\mu_{A2} - \mu_{A1} = (\mu_{21} + \mu_{22})/2 - (\mu_{12} + \mu_{11})/2.$$

Similarly, the main-effect of factor B is

$$\mu_{B2} - \mu_{B1} = (\mu_{12} + \mu_{22})/2 - (\mu_{11} + \mu_{21})/2.$$

§2.6. Variation components in two-way ANOVA

- In the data of a two-way ANOVA model, variation components include those due to the main effects of the two factors, (SSA and SSB), the interaction (SSAB) and the error (SSE). The total sum of squares (SST) (total variation) is decomposed as the sum of those components.
- The sum of squares measuring the variations are given by

$$\mathrm{SST} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n_{ij}}(y_{ijk} - \bar{y}_{\cdots})^2, \ \mathrm{SSA} = \sum_{i=1}^{a} n_{i\cdot}(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2,$$

$$\mathrm{SSB} = \sum_{j=1}^{b} n_{\cdot j}(\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2, \ \mathrm{SSAB} = \sum_{i=1}^{a}\sum_{j=1}^{b} n_{ij}(\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2,$$

$$\mathrm{SSE} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n_{ij}}(y_{ijk} - \bar{y}_{ij\cdot})^2, \quad \bar{y}_{i\cdot\cdot} = \frac{1}{n_{i\cdot}}\sum_{j=1}^{b}\sum_{k=1}^{n_{ij}} y_{ijk}, \ n_{i\cdot} = \sum_{j=1}^{b} n_{ij},$$

$$\bar{y}_{\cdot j\cdot} = \frac{1}{n_{\cdot j}}\sum_{i=1}^{a}\sum_{k=1}^{n_{ij}} y_{ijk}, \ n_{\cdot j} = \sum_{i=1}^{I} n_{ij}, \ \bar{y}_{\cdots} = \frac{1}{n}\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n_{ij}} y_{ijk}, \quad n = \sum_{i=1}^{a} n_{i\cdot},$$

$$\bar{y}_{ij\cdot} = \frac{1}{n_{ij}}\sum_{k=1}^{n_{ij}} y_{ijk}.$$

§2.7. Two-way ANOVA table

- The two-way ANOVA table is given by

Sourse	d.f.	SS	MS	F-test
A	a-1	SSA	MSA	MSA/MSE
B	b-1	SSB	MSB	MSB/MSE
AB	(a-1)(b-1)	SSAB	MSAB	MSAB/MSE
Error	n-ab	SSE	MSE	–
Total	n-1	SST	–	–

- Test for interaction: $F = \text{MSAB}/\text{MSE}$. Under the hypothesis of no interaction, $F \sim F_{(a-1)(b-1),n-ab}$.
- Test for main effect: $F_1 = \text{MSA}/\text{MSE}$, $F_2 = \text{MSB}/\text{MSE}$. Under hypothesis of zero main effects, $F_1 \sim F_{a-1,n-ab}$, $F_2 \sim F_{b-1,n-ab}$.
- Note: A factor has zero main effect does not imply that the factor has no effect, unless the interaction effect does not exist.

§2.8. Human Relations Course (HRC) Example

- New police officers of a US police department in a human relations course are assigned to 9 groups according to Factor A: type of beat to which officers are assigned during the course, and Factor B: the length of the course.
- Factor A has three levels: upper class beat, middle class beat and inner-city beat.
- Factor B has three levels: five hours, ten hours and fifteen hours.
- The assessment measure is a test score on the attitude towards minority groups following the course.
- The data is given in the file `HRCdata.txt` on Canvas and is presented on the next slide.

§2.9. HRC example: data

Factor A	Factor B		
	b_1	b_2	b_3
a_1	24	44	38
	33	36	29
	37	25	28
	29	27	47
	42	43	48
a_2	30	35	26
	21	40	27
	39	27	36
	26	31	46
	34	22	45
a_3	21	41	42
	18	39	52
	10	50	53
	31	36	49
	20	34	64

§2.10. HRC example, ANOVA

- ANOVA table

Source	Df	Sum Sq	Mean Sq	F value	p-value
Beat	2	190.0	95.00	1.5200	0.23242
Time	2	1543.3	771.67	12.3467	0.00008
Beat.Time	4	1236.7	309.17	4.9467	0.00278
Residuals	36	2250.0	62.50		

- The p-values are computed by:

```
pf(1.52,2,36,lower.tail=FALSE)
pf(12.3467,2,36,lower.tail=FALSE)
pf(4.9467,4,36,lower.tail=FALSE)
```

- Conclusion: The type of Beat and course Time have a significant interaction effect on the attitude of the officers; the main effect of Time is significant, but the main effect of Beat is not.

§2.11. Issues with the traditional approach

- For the traditional approach to be valid, it requires that the data is balanced, that is, the number of observations n_{ij} are the same at each combination of the levels.
- If the data is not balanced, the SSAB does not measure the interaction effect, SSA and SSB do not measure purely the main effects.
- The above points are illustrate by a two-way ANOVA model while both factors have two levels:

		B		Level
		1	2	mean of A
A	1	μ_{11}	μ_{12}	$\mu_{1\cdot}$
	2	μ_{21}	μ_{22}	$\mu_{2\cdot}$
level mean of B		$\mu_{\cdot 1}$	$\mu_{\cdot 2}$	$\mu_{\cdot\cdot}$

- Effect of A at two levels of B:

$$\mu_{11} - \mu_{21}, \quad \mu_{12} - \mu_{22}.$$

- Interaction effect:

$$\mu_{11} - \mu_{21} - \mu_{12} + \mu_{22}.$$

§2.12. Issues with the interaction (1)

In the term $(\bar{y}_{11\cdot} - \bar{y}_{1\cdot\cdot} - \bar{y}_{\cdot 1\cdot} + \bar{y}_{\cdot\cdot\cdot})^2$ of SSAB:

$$
\begin{aligned}
& \bar{y}_{11\cdot} - \bar{y}_{1\cdot\cdot} - \bar{y}_{\cdot 1\cdot} + \bar{y}_{\cdot\cdot\cdot} \\
= \quad & \bar{y}_{11\cdot} - \frac{n_{11}\bar{y}_{11} + n_{12}\bar{y}_{12}}{n_{11} + n_{12}} - \frac{n_{11}\bar{y}_{11\cdot} + n_{21}\bar{y}_{21\cdot}}{n_{11} + n_{21}} \\
& + \frac{n_{11}\bar{y}_{11\cdot} + n_{21}\bar{y}_{21\cdot} + n_{12}\bar{y}_{12\cdot} + n_{22}\bar{y}_{22\cdot}}{n_{11} + n_{21} + n_{12} + n_{22}}. \\
= \quad & \left(1 - \frac{n_{11}}{n_{11} + n_{12}} - \frac{n_{11}}{n_{11} + n_{21}} + \frac{n_{11}}{n_{11} + n_{21} + n_{12} + n_{22}}\right) \bar{y}_{11\cdot} \\
& - \left(\frac{n_{21}}{n_{11} + n_{21}} - \frac{n_{21}}{n_{11} + n_{21} + n_{12} + n_{22}}\right) \bar{y}_{21\cdot} \\
& - \left(\frac{n_{12}}{n_{11} + n_{12}} - \frac{n_{12}}{n_{11} + n_{21} + n_{12} + n_{22}}\right) \bar{y}_{12\cdot} \\
& + \frac{n_{22}}{n_{11} + n_{21} + n_{12} + n_{22}} \bar{y}_{22\cdot} \\
\equiv \quad & c_{11}\bar{y}_{11\cdot} - c_{21}\bar{y}_{21\cdot} - c_{12}\bar{y}_{12\cdot} + c_{22}\bar{y}_{22\cdot}
\end{aligned}
$$

§2.13. Issues with the interaction (2)

- The expectation of $\bar{y}_{11\cdot} - \bar{y}_{1\cdot\cdot} - \bar{y}_{\cdot 1\cdot} + \bar{y}_{\cdot\cdot\cdot}$ is

$$c_{11}\mu_{11} - c_{21}\mu_{21} - c_{12}\mu_{12} + c_{22}\mu_{22}.$$

  If the c_{ij}'s are not equal, it is not a measure of the interaction effect.
- Only when $n_{11} = \cdots = n_{22} = n$, it reduces to

$$\frac{1}{4}(\mu_{11} - \mu_{21} - \mu_{12} + \mu_{22}).$$

  That is, only in this case, SSAB measures the interaction effect.
- If the n_{ij}'s are different, SSAB does not provide any meaningful measure on the interaction effect.

§2.14. Issues with main effect (1)

In the term $(\bar{y}_{1\cdot\cdot} - \bar{y}_{\cdots})^2$ of SSA:

$$
\begin{aligned}
& \bar{y}_{1\cdot\cdot} - \bar{y}_{\cdots} \\
= \quad & \frac{n_{11\cdot}\bar{y}_{11\cdot} + n_{12}\bar{y}_{12\cdot}}{n_{11} + n_{12}} - \frac{n_{11}\bar{y}_{11\cdot} + n_{21}\bar{y}_{21\cdot} + n_{12}\bar{y}_{12\cdot} + n_{22}\bar{y}_{22\cdot}}{n_{11} + n_{21} + n_{12} + n_{22}}. \\
= \quad & \left(\frac{n_{11}}{n_{11} + n_{12}} - \frac{n_{11}}{n_{11} + n_{21} + n_{12} + n_{22}} \right) \bar{y}_{11\cdot} \\
& + \left(\frac{n_{12}}{n_{11} + n_{12}} - \frac{n_{12}}{n_{11} + n_{21} + n_{12} + n_{22}} \right) \bar{y}_{12\cdot} \\
& - \frac{n_{21}}{n_{11} + n_{21} + n_{12} + n_{22}} \bar{y}_{21\cdot} \\
& - \frac{n_{22}}{n_{11} + n_{21} + n_{12} + n_{22}} \bar{y}_{22\cdot} \\
\equiv \quad & c_{11}\bar{y}_{11\cdot} - c_{21}\bar{y}_{21\cdot} + c_{12}\bar{y}_{12\cdot} - c_{22}\bar{y}_{22\cdot}
\end{aligned}
$$

§2.15. Issues with main effect (2)

- The expectation of $\bar{y}_{1\cdot\cdot} - \bar{y}_{\cdots}$ is

$$\begin{aligned} & c_{11}\mu_{11} - c_{21}\mu_{21} + c_{12}\mu_{12} - c_{22}\mu_{22} \\ = \quad & c_{11}(\mu_{11} - \mu_{21}) + (c_{11} - c_{21})\mu_{21} + c_{12}(\mu_{12} - \mu_{22}) + (c_{12} - c_{22})\mu_{22}, \\ = \quad & c_{11}(\mu_{11} - \mu_{21}) + c_{12}(\mu_{12} - \mu_{22}) + (c_{11} - c_{21})(\mu_{21} - \mu_{22}) \\ & +(c_{11} - c_{21} + c_{12} - c_{22})\mu_{22}. \end{aligned}$$

- When $n_{11} = \cdots = n_{22} = n$, it reduces to

$$\frac{1}{4}(\mu_{11} - \mu_{21}) + \frac{1}{4}(\mu_{12} - \mu_{22}),$$

  which measures the effect of A only.
- If the n_{ij}'s are different, only the first two terms measure the effect of A, the other terms measure the effect of B. The effects of A and B are confounded in $(\bar{y}_{1\cdot\cdot} - \bar{y}_{\cdots})^2$.

§3.1. Dummy variables

- A factor can be represented by dummy variables. The one-way and two-way ANOVA problems can be studied by regression models with the dummy variables as covariates.
- If the factor has a levels, it can be represented by $a-1$ dummy variables.
- The conventional definition of dummy variables:

$$u_k = \begin{cases} 1, & \text{if level } k; \\ 0, & \text{otherwise,} \end{cases}$$
$$k = 2, \ldots, a.$$

- Definition of dummy variables is not unique. In regression models with factor predictors, for different definition of dummy variables, the regression coefficients have different meanings.

§3.2. LRM for one-way ANOVA

- The one-way ANOVA model can be expressed using the dummy variables u_k as

$$y_{ij} = \beta_0 + \beta_2 u_{i2} + \cdots + \beta_a u_{ia} + \epsilon_{ij}.$$

- In matrix form: $\boldsymbol{y} = X\beta + \boldsymbol{\epsilon}$, where

$$\boldsymbol{y} = (y_{11}, \ldots, y_{1n_1}, \cdots, y_{a1}, \ldots, y_{an_a})^\top,$$
$$\boldsymbol{\epsilon} = (\epsilon_{11}, \ldots, \epsilon_{1n_1}, \cdots, \epsilon_{a1}, \ldots, \epsilon_{an_a})^\top,$$
$$X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{1}_{n_2} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{1}_{n_a} & \mathbf{0} & \cdots & \mathbf{1}_{n_a} \end{pmatrix}.$$

§3.3. Interpretation of the regression parameters

- At different levels of the factor, the model is expressed separately as

$$y_{ij} = \begin{cases} \beta_0 + \epsilon_{1j}, & \text{if level 1 } (i.e., \text{i=1}); \\ \beta_0 + \beta_k + \epsilon_{kj}, & \text{if level } k \ (i.e., \text{i=k}), \ k = 2, \ldots, a. \end{cases}$$

- Taking expectation on both sides of the models, the relation between regression parameters and the mean response at each level is obtained as:

$$\mu_1 = \beta_0, \ \mu_k = \beta_0 + \beta_k, \ k = 2, \ldots, a.$$

- The parameters have the following meaning: β_0 is the expected response at level 1, $\beta_k = \mu_k - \mu_1$ is the difference between the expected responses at level k and level 1.
- $\beta_k, k = 2, \ldots, a$ are $a - 1$ independent contrasts of the level means. Any contrast of the level means can be expressed as a linear combination of these $a - 1$ β_k's.

$$\sum_{i=1}^{a} c_i \mu_i = \sum_{i=2}^{a} c_i \beta_i.$$

§3.4. Inference with regression approach

- The significance of the factor effect is tested by

$$F = \frac{MSR}{MSE}.$$

  Under the hypothesis that the factor has no effect, $F \sim F_{a-1,n-a}$.

- The contrast hypothesis $\sum_i c_i \mu_i = 0$ is tested by testing the linear hypothesis $\sum_{i=2}^{a} c_i \beta_i = 0$. The test statistic is given by

$$T = \boldsymbol{c}_1^\top \hat{\boldsymbol{\beta}}_1 / \sqrt{\boldsymbol{c}_1^\top \hat{\Sigma} \boldsymbol{c}_1},$$

  where $\boldsymbol{c}_1 = (c_2, \ldots, c_a)^\top$, $\hat{\boldsymbol{\beta}}_1 = (\hat{\beta}_2, \ldots, \hat{\beta}_a)^\top$ and $\hat{\Sigma}$ is the corresponding variance matrix.

- Confidence intervals of $\sum_{i=1}^{a} c_i \mu_i$ is replaced by the confidence intervals of $\sum_{i=2}^{a} c_i \beta_i$.

§3.5. KFC exmaple (§2.3): regression approach

- Generate the vectors:

```
y=c(11,17,16,14,15,12,10,15,19,11,23,20,18,
   17,27,33,22,26,28)
u2=c(0,0,0,0,0,1,1,1,1,1,0,0,0,0,0,0,0,0,0)
u3=c(0,0,0,0,0,0,0,0,0,0,1,1,1,1,0,0,0,0,0)
u4=c(0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,1,1,1)
```

- Fit the model:

```
reg.fit = lm(y~u2+u3+u4)

summary(reg.fit)

anova(reg.fit)
```

§3.6. KFC exmaple: fitted results

- Estimates:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.600      1.452  10.053 4.66e-08
u2            -1.200      2.054  -0.584   0.5677
u3             4.900      2.179   2.249   0.0399
u4            12.600      2.054   6.135 1.91e-05
```

- ANOVA table:

```
          Df Sum Sq Mean Sq F value    Pr(>F)
u2         1 185.72  185.72  17.610 0.0007786
u3         1   5.60    5.60   0.531 0.4774213
u4         1 396.90  396.90  37.633  1.91e-05
Residuals 15 158.20   10.55
```

The table can be collapsed to the previously obtained ANOVA table:

Source	df	SS	MS	F
Factor A	3	588.22	196.07	18.6
Error	15	158.20	10.55	
Total	18	746.42		

§3.7. The R function lm with factor arguments

- By using R, we don't need to generate the dummy variables explicitly. The R function `lm` generates automatically the conventional dummy variables for a factor.
- In data preparation, we only need to create the response vector and the factor vector, e.g.,

```
y=c(11,17,16,14,15,12,10,15,19,11,23,20,18,17,27,33,22,26,28)
package=factor(c(1,1,1,1,1,2,2,2,2,2,3,3,3,3,4,4,4,4,4))
```

- The formula within `lm` is specified in terms of the factor, e.g., `reg.fit = lm(y~package)`. In the example, it yields the results:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.600      1.452  10.053 4.66e-08 ***
package2      -1.200      2.054  -0.584   0.5677
package3       4.900      2.179   2.249   0.0399 *
package4      12.600      2.054   6.135 1.91e-05 ***
Analysis of Variance Table:
          Df Sum Sq Mean Sq F value    Pr(>F)
package    3 588.22 196.074  18.591 2.585e-05 ***
Residuals 15 158.20  10.547
Variance matrix:
            (Intercept)  package2  package3  package4
(Intercept)    2.109333 -2.109333 -2.109333 -2.109333
package2      -2.109333  4.218667  2.109333  2.109333
package3      -2.109333  2.109333  4.746000  2.109333
package4      -2.109333  2.109333  2.109333  4.218667
```

§3.8. KFC example: classroom questions

Given the results in the previous slide,

(i) Test whether package design has a significant effect on the sales at level $\alpha = 0.01$.

(ii) Construct a 95% confidence interval for the expected sales with package design 1 and package design 2.

(iii) Test at level $\alpha = 0.05$ whether each of package designs 2, 3 and 4 is significantly different from package design 1.

(iv) Test at level $\alpha = 0.05$ whether package design 2 and package design 4 are significantly different.

Denote the regression model by

$$y = \beta_0 + \beta_2 pk2 + \beta_3 pk3 + \beta_4 pk4 + \epsilon.$$

Let μ_j denote the expected sales with package design j.

§3.9. KFC example: solution to classroom questions (1)

(i) The test statistic for the significant test $F = 18.591$ has a p-value 2.5853−05 which is smaller than 0.01. It is concluded that package design has a significant effect on the sales.

(ii) The expected sales with package design 1 and package design 2 are $\mu_1 = \beta_0$ and $\mu_2 = \beta_2 + \beta_0$. The c.i. for β_0 is given by

$$\begin{aligned} & \hat{\beta}_0 \pm sd(\hat{\beta}_0) t_{15}(0.025) \\ = \quad & 14.6 \pm 1.452 * 2.13145 = [11.5051,\ 17.6949]. \end{aligned}$$

The c.i. for $\beta_2 + \beta_0$ is given by

$$\begin{aligned} & \hat{\beta}_2 + \hat{\beta}_0 \pm sd(\hat{\beta}_2 + \hat{\beta}_0) t_{15}(0.025) \\ = \quad & 13.4 \pm 1.4528 * 2.13145 = [10.3034,\ 16.4966]. \end{aligned}$$

where

$$sd(\hat{\beta}_2 + \hat{\beta}_0) = \sqrt{4.2187 + 2.1093 - 2 * 2.1093} = 1.4528.$$

§3.10. KFC example: solution to classroom questions (2)

(iii) The hypotheses of the tests are: $H_0 : \beta_j = 0, vs. H_1 : \beta_j \neq 0, j = 2, 3, 4$. The values of the test statistics are given in the summary table under the heading `t value`. The p-value of the tests are given under the heading `Pr(>|t|)`. For package 2, the p-value $0.5677 > 0.05$, H_0 is not rejected; for package designs 3 and 4, the p-values 0.0399 and `1.91e-05` are smaller than 0.05, H_0 is rejected. It is concluded that no strong evidence to support a significant difference between package design 2 and package design 1, package design 3 and 4 have significant difference from package design 1.

(iv) It amounts to test $H_0 : \beta_2 - \beta_4 = 0, v.s. H_1 : \beta_2 - \beta_4 \neq 0$. The test statistic

$$\begin{aligned} T &= (\hat{\beta}_2 - \hat{\beta}_4)/sd(\hat{\beta}_2 - \hat{\beta}_4) \\ &= (-1.2 - 12.6)/\sqrt{4.2187 + 4.2187 - 2 * 1.1093} = -5.5338. \end{aligned}$$

Since $|T| = 5.5338 > 2.13145 = t_{15}(0.025)$, H_0 is rejected. It is concluded that package design 2 and 4 are significantly different.

§3.11. Regression models for two-way ANOVA

- Two way ANOVA data involves two factors: A with a levels and B with b levels. Each factor is represented by a set of dummy variables. The dummy variables are defined in the same way as that for one-way ANOVA models:

$$u_i = \begin{cases} 1, & \text{if level } i; \\ 0, & \text{otherwise,} \end{cases} \quad i = 2, \ldots, a;$$

$$v_j = \begin{cases} 1, & \text{if level } j; \\ 0, & \text{otherwise,} \end{cases} \quad j = 2, \ldots, b.$$

- For two-way ANOVA data, two models can be considered: main-effect model and interaction model. Interaction model can be considered only when there are more than one observations at the level combinations (cells).
- The main-effect model is given by

$$y = \beta_0 + \sum_{i=2}^{a} \alpha_{\mathrm{m}i} u_i + \sum_{j=2}^{b} \beta_{\mathrm{m}j} v_j + \epsilon.$$

§3.12. The parameter's meaning in the main effect model

- The expectations of the main effect model at different levels are as follows.

$$Ey = \begin{cases} \mu_{11} = \beta_0, & \text{both } A \text{ and } B \text{ at level 1,} \\ \mu_{i1} = \beta_0 + \alpha_{\mathrm{m}i}, & A \text{ at level } i, B \text{ at level 1,} \\ \mu_{1j} = \beta_0 + \beta_{\mathrm{m}j}, & A \text{ at level 1}, B \text{ at level } j, \\ \mu_{ij} = \beta_0 + \alpha_{\mathrm{m}i} + \beta_{\mathrm{m}j}, & A \text{ at level } i, B \text{ at level } j. \end{cases}$$

- Interpretation:
  - $\alpha_{\mathrm{m}i} = \mu_{i1} - \mu_{11}$: the effect of A when its level changes from 1 to i while B is fixed at level 1, which is the same as $\mu_{ij} - \mu_{1j}$, the same effect of A while B is fixed at level j. The effect does not depend on the level of B.
  - $\beta_{\mathrm{m}j} = \mu_{1j} - \mu_{11}$, the effect of B when its level changes from 1 to j while A is fixed at level 1, which is the same as $\mu_{ij} - \mu_{i1}$, the same effect of B while A is fixed at level i. The effect does not depend on the level of A.
- In the main-effect model, the effect of one factor is not affected by the effect of the other factor.

§3.13. Interaction model

- Interaction model is given by
$$y = \beta_0 + \sum_{i=2}^{a} \alpha_i u_i + \sum_{j=2}^{b} \beta_j v_j + \sum_{i=2}^{a} \sum_{j=2}^{b} \xi_{ij} u_i v_j + \epsilon.$$
- The expectation of the interaction model at different levels:
$$Ey_{ijk} = \begin{cases} \mu_{11} = \beta_0, & \text{both } A \text{ and } B \text{ at level } 1, \\ \mu_{i1} = \beta_0 + \alpha_i, & A \text{ at level } i, B \text{ at level } 1, \\ \mu_{1j} = \beta_0 + \beta_j, & A \text{ at level } 1, B \text{ at level } j, \\ \mu_{ij} = \beta_0 + \alpha_i + \beta_j + \xi_{ij}, & A \text{ at level } i, B \text{ at level } j. \end{cases}$$
  - $\alpha_i = \mu_{i1} - \mu_{11}$: the effect of A when its level changes from 1 to i while B is fixed at level 1.
  - $\beta_j = \mu_{1j} - \mu_{11}$: the effect of B when its level changes from 1 to j while A is fixed at level 1.
  - $\alpha_i + \xi_{ij} = \mu_{ij} - \mu_{1j}$: the effect of A when its level changes from 1 to i while B is fixed at level j.
  - $\beta_j + \xi_{ij} = \mu_{ij} - \mu_{i1}$: the effect of B when its level changes from 1 to j while A is fixed at level i.
  - $\xi_{ij} = (\mu_{ij} - \mu_{i1}) - (\mu_{1j} - \mu_{11})$, an interaction contrast.

§3.14. Main effect and local effect

- In the main effect model, the effect of one factor is not affected by the other factor, it is the overall effect of that factor ignoring the effect of the other factor. This overall effect is referred to as the main effect. E.g., $\alpha_{\mathrm{m}i}$ is the main-effect of A when its level changes for 1 to i; $\beta_{\mathrm{m}j}$ is the main-effect of B when its level changes from 1 to j.
- In the interaction model, the effect of one factor differs at different levels of the other factor. The effect of a factor at a particular level of the other factor is its local effect. E.g., α_i is the local effect of A when its level changes from 1 to i while B is fixed at level 1, β_j is the local effect of B when its level changes from 1 to j while A is fixed at level 1.
- In the interaction model, the main effect of one factor is its corresponding local effects integrated over all the levels of the other factor. If the data is balanced, the main effect is the average of the local effects, e.g., $\alpha_{\mathrm{m}i} = \alpha_i + (1/b)\sum_{j=2}^{b} \xi_{ij}$. But, in general, the main effect is not simply the average of the local effects.

§3.15. HRC example (§2.8) (balanced data)

- Estimated coefficients of interaction model:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   33.000      3.535   9.334 3.79e-11 ***
beat2         -3.000      5.000  -0.600  0.55226
beat3        -13.000      5.000  -2.600  0.01344 *
time2          2.000      5.000   0.400  0.69152
time3          5.000      5.000   1.000  0.32399
beat2:time2   -1.000      7.071  -0.141  0.88833
beat3:time2   18.000      7.071   2.546  0.01534 *
beat2:time3    1.000      7.071   0.141  0.88833
beat3:time3   27.000      7.071   3.818  0.00051 *
```

- Estimated coefficients of main-effect model:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   28.000      3.112   8.997 3.71e-11 ***
beat2         -3.000      3.409  -0.880 0.384122
beat3          2.000      3.409   0.587 0.560730
time2          7.667      3.409   2.249 0.030097 *
time3         14.333      3.409   4.204 0.000143 ***
```

- Main effect contrast: $\hat{\alpha}_{m3} = 2$.
  Local contrasts: $\hat{\alpha}_3 = -13$, $\hat{\alpha}_3 + \hat{\xi}_{32} = -13 + 18 = 5$, $\hat{\alpha}_3 + \hat{\xi}_{33} = -13 + 27 = 14$, and the average of the local contrasts equal to $(-13 + 5 + 14)/3 = 2$, i.e.,

$$\hat{\alpha}_{m3} = \hat{\alpha}_3 + \frac{1}{3}(\hat{\xi}_{32} + \hat{\xi}_{33}).$$

§3.16. DVD example (unbalanced data)

- Estimated coefficients of interaction model:

```
Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)               3.2664     0.2514  12.991   <2e-16 ***
RatingPG-13               0.2362     0.3496   0.675    0.501
RatingR                   0.1194     0.4591   0.260    0.795
GenreComedy               0.1294     0.3895   0.332    0.740
GenreDrama               -0.5696     0.7112  -0.801    0.425
RatingPG-13:GenreComedy  -0.8192     0.5127  -1.598    0.113
RatingR:GenreComedy      -0.4514     0.6162  -0.733    0.465
RatingPG-13:GenreDrama   -0.1053     0.7757  -0.136    0.892
RatingR:GenreDrama       -0.4961     0.8366  -0.593    0.554
```

- Estimated coefficients of main-effect model:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.44981    0.20887  16.517   <2e-16 ***
RatingPG-13 -0.04356    0.23521  -0.185   0.8534
RatingR     -0.25118    0.26058  -0.964   0.3371
GenreComedy -0.31876    0.22365  -1.425   0.1568
GenreDrama  -0.71296    0.22861  -3.119   0.0023 **
```

- Main effect contrast: $\hat{\alpha}_{m3} = -0.25118$.
  Note that $\hat{\alpha}_3 = 0.1194$, $\hat{\xi}_{32} = -0.4514$, $\hat{\xi}_{33} = -0.4961$, but

$$\hat{\alpha}_3 + \frac{1}{3}(\hat{\xi}_{32} + \hat{\xi}_{33}) = 0.1194 - (0.4514 + 0.4961)/3 = -0.1964333 \neq -0.25118.$$

§3.17. Inference on overall interaction effect

- For testing whether there is a significant overall interaction effect, under the regression model, it is equivalent to test

$$H_0 : \xi_{ij} = 0, i = 2, \ldots, 1; j = 2, \ldots, b;$$
$$\text{vs.} \quad H_1 : \text{at least one of } \xi_{ij} \neq 0,$$

  since ξ_{ij}'s are the basis interaction contrasts.
- It is tested by a F-test. The F-test statistic can be obtained
  - either from the table produced by `anova`,
  - or through the computation of Wald statistic as follows: let $\hat{\boldsymbol{\xi}}$ be the vector of estimated ξ_{kj}'s and $\hat{\Sigma}_\xi$ the estimated covariance matrix of $\hat{\boldsymbol{\xi}}$. First compute the Wald statistic

$$W = \hat{\boldsymbol{\xi}}^\top \hat{\Sigma}_\xi^{-1} \hat{\boldsymbol{\xi}}.$$

    The F-statistic is given by $F = W/[(a-1)(b-1)]$
- Under H_0, $F \sim F_{(a-1)(b-1),n-ab}$. The test reject H_0 at level α, if $F > F_{(a-1)(b-1),n-ab}(\alpha)$, or equivalently, the p-value `pf(F,(a-1)(b-1), n-ab,lower.tail=FALSE)` $< \alpha$.

§3.18. Inference on particular interaction effects

- In addition to the overall interaction effect, particular interaction effects can be investigated through interaction contrasts which are linear combinations of ξ_{ij}'s.
- Examples:
  - Test whether the effect of factor B when its level changes from 1 to j is the same at level i and 1 of Factor A:

$$H_0 : (\mu_{ij} - \mu_{i1}) - (\mu_{1j} - \mu_{11}) = \xi_{ij} = 0.$$

  - Test whether the effect of factor B when its level changes from ℓ to j is the same at level i and 1 of Factor A:

$$H_0 : (\mu_{ij} - \mu_{i\ell}) - (\mu_{1j} - \mu_{1\ell}) = \xi_{ij} - \xi_{i\ell} = 0.$$

  - Test whether the effect of factor B when its level changes from k to j is the same at level i and ℓ of Factor A:

$$H_0 : (\mu_{ij} - \mu_{ik}) - (\mu_{\ell j} - \mu_{\ell k}) = \xi_{ij} - \xi_{ik} - \xi_{\ell j} + \xi_{\ell k} = 0.$$

§3.19. Inference on particular interaction effects (cont.)

- Any interaction contrast can be expressed in the form $\boldsymbol{c}^\top \boldsymbol{\xi}$, where $\boldsymbol{\xi}$ is the vector of ξ_{kj}'s. Let $\hat{\boldsymbol{\xi}}$ be the vector of estimated ξ_{kj}'s and $\hat{\Sigma}_\xi$ the estimated covariance matrix of $\hat{\boldsymbol{\xi}}$. The test statistic for $\boldsymbol{c}^\top \boldsymbol{\xi} = 0$ is given by
$$T_{\boldsymbol{c}} = \frac{\boldsymbol{c}^\top \hat{\boldsymbol{\xi}}}{\sqrt{\boldsymbol{c}^\top \hat{\Sigma}_\xi \boldsymbol{c}}}.$$
- Under $H_0 : \boldsymbol{c}^\top \boldsymbol{\xi} = 0$, $T_{\boldsymbol{c}} \sim t_{n-ab}$.
- For a particular interaction contrast, the above formula can be simplified. For example, to test $H_0 : \xi_{kj} - \xi_{kl} - \xi_{mj} + \xi_{ml} = 0$, only the vector $\hat{\boldsymbol{\theta}} = (\hat{\xi}_{kj}, \hat{\xi}_{kl}, \hat{\xi}_{mj}, \hat{\xi}_{ml})^\top$ and its covariance matrix $\hat{\Sigma}_\theta$ are needed, and the corresponding $\boldsymbol{c}$ reduces to its sub vector $\boldsymbol{b}$ with non-zero components, i.e., $\boldsymbol{b} = (1, -1, -1, 1)^\top$.
- The estimated sub vectors and the corresponding estimated covariance matrix can be obtained from the outputs of the `lm` function.

§3.20. HRC example: R codes

- R codes

```
HRC.dat = read.table("D.//Rsession/HRCdata.txt", header = TRUE)
B=factor(HRC.dat$beat)
T=factor(HRC.dat$time)
y=HRC.dat$y
HRC.int = lm(y~B*T)
summary(HRC.int)
anova(HRC.int)
b=HRC.int$coef;  V=vcov(HRC.int)

b.x=b[6:9]
v.x=V[6:9,6:9]
W=t(b.x)%*%solve(v.x)%*%b.x
F=W/4
```

- The first portion of the codes produces the results needed for inference. The second portion demonstrates the computation of the F-statistic for the overall interaction effect through Wald statistic.

§3.21. HRC example: fitted results

- Summary table:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   33.000      3.535   9.334 3.79e-11 ***
B2            -3.000      5.000  -0.600  0.55226
B3           -13.000      5.000  -2.600  0.01344 *
T2             2.000      5.000   0.400  0.69152
T3             5.000      5.000   1.000  0.32399
B2:T2         -1.000      7.071  -0.141  0.88833
B3:T2         18.000      7.071   2.546  0.01534 *
B2:T3          1.000      7.071   0.141  0.88833
B3:T3         27.000      7.071   3.818  0.00051 ***
```

- ANOVA table:

```
          Df Sum Sq Mean Sq F value    Pr(>F)
B          2  190.0   95.00  1.5200  0.232417
T          2 1543.3  771.67 12.3467 8.259e-05 ***
B:T        4 1236.7  309.17  4.9467  0.002781 **
Residuals 36 2250.0   62.50
```

- Variance matrix:

```
            (Intercept)    B2    B3    T2    T3 B2:T2 B3:T2 B2:T3 B3:T3
(Intercept)        12.5 -12.5 -12.5 -12.5 -12.5  12.5  12.5  12.5  12.5
B2                -12.5  25.0  12.5  12.5  12.5 -25.0 -12.5 -25.0 -12.5
B3                -12.5  12.5  25.0  12.5  12.5 -12.5 -25.0 -12.5 -25.0
T2                -12.5  12.5  12.5  25.0  12.5 -25.0 -25.0 -12.5 -12.5
T3                -12.5  12.5  12.5  12.5  25.0 -12.5 -12.5 -25.0 -25.0
B2:T2              12.5 -25.0 -12.5 -25.0 -12.5  50.0  25.0  25.0  12.5
B3:T2              12.5 -12.5 -25.0 -25.0 -12.5  25.0  50.0  12.5  25.0
B2:T3              12.5 -25.0 -12.5 -12.5 -25.0  25.0  12.5  50.0  25.0
B3:T3              12.5 -12.5 -25.0 -12.5 -25.0  12.5  25.0  25.0  50.0
```

§3.22. HRC example: Classroom questions I

Denote the model by

$$y = \beta_0 + \sum_{i=2}^{3} \alpha_i B_i + \sum_{j=2}^{3} \beta_j T_j + \sum_{i=2}^{3} \sum_{j=2}^{3} \xi_{ij} B_i T_j + \epsilon.$$

(i) Draw conclusion on whether there is any significant interaction effect between Time and Beat.

(ii) Interpret the meaning of $\xi_{ij}, i, j = 2, 3$. For each (i, j), at level 0.05, test

$$H_0 : \xi_{ij} = 0, v.s. H_1 : \xi_{ij} \neq 0.$$

(iii) Test whether the effect of time when it changes from ten hours to fifteen hours is significantly different between the beat level 3 and beat level 2.

Solution:

(i) The F test for the interaction effect has F-value 4.9467 and p-value 0.002781. With this small p-value, we can reject the null hypothesis of no interaction and conclude that the interaction effect between Time and Beat is significant.

§3.23. HRC example: Classroom questions I (cont.)

(ii) $\xi_{ij} = (\mu_{ij} - \mu_{i1}) - (\mu_{1j} - \mu_{11})$. It is the difference of the time effect when time changes from level 1 (5 hours) to level j (j=2, 10 hours, j=3, 15 hours) between beat level 1 (upper class) and beat level i (i=2, middle class, i=3, inner-city). The p-values of the tests can be obtained from the summary table, at level 0.05, ξ_{32} and ξ_{33} having, respectively, p-values 0.0153 and 0.0005 are significantly nonzero. ξ_{22} and ξ_{23} are not significantly nonzero.

(iii) At beat level 3 (inner-city), the effect when time changes from 10 hours to 15 hours is $\mu_{33} - \mu_{32}$, at beat level 2 (middle class), the effect is $\mu_{23} - \mu_{22}$. The difference of the effects is $\mu_{33} - \mu_{32} - \mu_{23} + \mu_{22}$. From the relation between the cell means and the regression parameters given on page §3.13, we have $\mu_{33} - \mu_{32} - \mu_{23} + \mu_{22} = \xi_{33} - \xi_{23} - \xi_{32} + \xi_{22}$. Thus, it amounts to test $H_0 : \xi_{33} - \xi_{23} - \xi_{32} + \xi_{22} = 0$.
Let $\boldsymbol{\xi} = (\xi_{22}, \xi_{32}, \xi_{23}, \xi_{33})^\top$, $\boldsymbol{c} = (1, -1, -1, 1)^\top$, $b.x = \hat{\boldsymbol{\xi}}$ and $v.x = \hat{\Sigma}_\xi$
The test statistic and p-value are computed by

```
c=c(1,-1,-1,1); T = t(c)%*%b.x/sqrt(t(c)%*%v.x%*%c)
2*pt(abs(T),36,lower.tail=FALSE)
```

yielding $T = 0.9899$, p-value $= 0.3288$. H_0 is not rejected.

§3.24. Sub-group study: inference on local effects

- If the interaction effect between the two factors is significant, the local effects of the factors should be analyzed. This kind of analysis is referred to in the literature as Sub-group study. The analysis of the local effects of factor A is discussed in detail as follows. The analysis for factor B is the same.
- At a fixed level of B, say level j, the overall significance of the effect of A is inferred by testing:

$$H_0 : \mu_{1j} = \mu_{2j} = \cdots = \mu_{aj}.$$

  The hypothesis is equivalent to

$$H_0 : \begin{cases} \alpha_2 = \cdots \alpha_a = 0, & \text{if } j = 1; \\ \alpha_2 + \xi_{2j} = \cdots = \alpha_a + \xi_{aj} = 0, & \text{if } j > 1. \end{cases}$$

- The local effect contrast $\sum_{i=1}^{a} c_i \mu_{ij}$ is equivalent to $\sum_{i=2}^{a} c_i \alpha_i$ if $j = 1$, and $\sum_{i=2}^{a} c_i(\alpha_i + \xi_{ij})$ if $j > 1$.

§3.25. Sub-group study: inference on local effects (cont.)

- To test either the overall significance or any contrasts, the hypothesis can be expressed in the form $H_0 : D\boldsymbol{\theta} = 0$, where D is either a matrix or a row vector, and $\boldsymbol{\theta}$ is the vector of the parameters involved in the hypothesis. E.g., for $\alpha_2 + \xi_{2j} = \cdots = \alpha_a + \xi_{aj} = 0$, we take $\boldsymbol{\theta} = (\boldsymbol{\alpha}^\top, \boldsymbol{\xi}_j^\top)^\top$ and $D = (I_{a-1}, I_{a-1})$, the hypothesis becomes $D\boldsymbol{\theta} = 0$.
- Let $\hat{\boldsymbol{\theta}}$ denote the estimate of $\boldsymbol{\theta}$, V denote the estimated variance matrix of $\hat{\boldsymbol{\theta}}$. The test statistic is given by

$$F = \hat{\boldsymbol{\theta}}^\top D^\top [DVD^\top]^{-1} D\hat{\boldsymbol{\theta}}/d,$$

  where d is the number of rows of D. Under H_0, the test statistic follows a F-distribution with d.f. d and $n - ab$ (the d.f. of error).
- If D is a row vector, say $\boldsymbol{c}^\top$, the F-statistic can be replaced by the t-statistic

$$T = \boldsymbol{c}^\top \hat{\boldsymbol{\theta}}/\sqrt{\boldsymbol{c}^\top V \boldsymbol{c}},$$

  which follows t_{n-ab} under H_0. Confidence intervals for $\boldsymbol{c}^\top \boldsymbol{\theta}$ can also be constructed.

§3.26. HRC example: Classroom questions II

(i) At level 0.05, test whether the training course length (Time) has a significant effect for inner-city beat.

(ii) At level 0.05, test whether the officers assigned to upper-class beat trained for 15 hours have better attitude than those trained for 10 hours.

Solution:

(i) The expected attitude scores with different training time for inner-city beat are $\mu_{33}, \mu_{32}, \mu_{31}$. It amounts to test

$$H_0 : \beta_2 + \xi_{32} = \beta_3 + \xi_{33} = 0.$$

Let $\boldsymbol{\theta} = (\beta_2, \beta_3, \xi_{32}, \xi_{33})^\top$ and $D = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$. From the fitted results, $\hat{\boldsymbol{\theta}} = (2, 5, 18, 27)^\top$, the corresponding variance matrix is

	T2	T3	B3:T2	B3:T3
T2	25.0	12.5	-25.0	-12.5
T3	12.5	25.0	-12.5	-25.0
B3:T2	-25.0	-12.5	50.0	25.0
B3:T3	-12.5	-25.0	25.0	50.0

§3.27. HRC example: Classroom questions II (cont.)

(i) (cont.) The test statistic $F = \hat{\boldsymbol{\theta}}^\top D^\top [DVD^\top]^{-1} D\hat{\boldsymbol{\theta}}/2$ is computed as $F = 20.9068$. The p-value is computed as `pf(F,2,36,lower.tail=FALSE)= 9.429152e-07`. It is concluded that the training course length has a significant effect on the officers assigned to inner-city beat.

(ii) The expected attitude scores for the officers assigned to upper-class beat trained for 15 hours and for 10 hours are μ_{13}, μ_{12}. It amounts to test

$$H_0 : \mu_{13} - \mu_{12} \leq 0 \ v.s. \ H_1 : \mu_{13} - \mu_{12} > 0$$

which is equivalent to

$$H_0 : \beta_3 - \beta_2 \leq 0 \ v.s. \ H_1 : \beta_3 - \beta_2 > 0.$$

The test statistic is computed as

$$\begin{aligned} T &= \frac{\hat{\beta}_3 - \hat{\beta}_2}{\sqrt{\mathrm{Var}(\hat{\beta}_3) + \mathrm{Var}(\hat{\beta}_2) - 2\mathrm{Cov}(\hat{\beta}_3, \hat{\beta}_2)}} \\ &= \frac{5-2}{\sqrt{25 + 25 - 2 * 12.5}} = 0.6. \end{aligned}$$

The p-value is computed as `1-pt(0.6,36)=0.2761323`. H_0 is not rejected.

§3.28. Inference on main-effects

- If the cells are balanced, i.e., the numbers of observation in the cells are the same, the significance of main effects can be tested using the ANONA table produced by the `anova` function from the interaction model.
- If the cells are unbalanced, the main effect SS's produced by `anova` depend on the order of the factors in the formula specification in `lm`. Only the last main effect SS is the correct one. If we want to get the correct SS and the corresponding F values for both factors, the model must be fitted twice, each time with a different order of the factors in the formula specification.

§3.29. HRC example: inference on main effect

- The data in the example is balanced.
- ANOVA table from the fitting with factor order: beat, time:

```
          Df Sum Sq Mean Sq F value    Pr(>F)
B          2  190.0   95.00  1.5200  0.232417
T          2 1543.3  771.67 12.3467 8.259e-05 ***
B:T        4 1236.7  309.17  4.9467  0.002781 **
Residuals 36 2250.0   62.50
```

- ANOVA table from the fitting with factor order: time, beat:

```
          Df Sum Sq Mean Sq F value    Pr(>F)
T          2 1543.3  771.67 12.3467 8.259e-05 ***
B          2  190.0   95.00  1.5200  0.232417
T:B        4 1236.7  309.17  4.9467  0.002781 **
Residuals 36 2250.0   62.50
```

- The main effect of Time is significant. But that of beat is not.

§3.30. Issues on inference for main-effect contrasts

- In the interaction model, the main effect contrasts have complicated structures which cannot be expressed in an explicit form of the regression parameters. Only when the data is balanced, a main-effect contrast can be expressed as the average of the local contrasts, e.g., $\alpha_{\mathrm{m}i} = \alpha_i + (1/b)\sum_{j=2}^{b} \xi_{ij}$.
- It is not convenient, if not impossible, to use the fitted results of the interaction model for inference on main effect contrast. Even in the balanced data case, it will be tedious.
- The inference on main effect contrasts can be based on the main effect model, which produces correct estimates for main-effect contrasts. But the estimated variance matrix of the main effect model must be adjusted.

§3.31. Adjustment of error variance estimate.

- Let X_m denote the design matrix of the main-effect model. The estimated variance matrix from the main-effect model is given by

$$V.m = \hat{\sigma}_M^2 (X_m^\top X_m)^{-1},$$

where $\hat{\sigma}_M^2$ is the MSE of the main-effect model which is not a proper estimate of σ^2. Since the SSE of the main-effect model contains not only the variation caused by errors but also those caused by the interactions. The proper estimate of σ^2 should be $\hat{\sigma}_I^2$, the MSE of the interaction model.

- The estimated variace matrix should be adjusted to

$$V = \hat{\sigma}_I^2 (X_m^\top X_m)^{-1} = (\hat{\sigma}_I^2 / \hat{\sigma}_m^2) V.m.$$

- The inference on main effect contrasts are carried out by using the estimates from the main-effect model and the adjusted estimated variance matrix. The d.f. of the t-statistics for testing main-effect contrast is the d.f. of the MSE in the interaction model.

§3.32. HRC example: adjustment of variance matrix of the main-effect model

- R codes for making adjustment of variance matrix and summary table:

```
# Adjustment for variance matrix
sigma.I = summary(HRC.int1)$sigma
HRC.main = lm(y~beat+time,data=HRC.dat); summ.m=summary(HRC.main)
V.m = vcov(HRC.main)     # variance matrix from main-effect model
sigma.M = summ.m$sigma
V = V.m*sigma.I^2/sigma.M^2  # adjusted variance matrix
```

- Un-adjusted variance matrix:

```
            (Intercept)      B2      B3      T2      T3
(Intercept)      9.6852 -5.8111 -5.8111 -5.8111 -5.8111
B2              -5.8111 11.6222  5.8111  0.0000  0.0000
B3              -5.8111  5.8111 11.6222  0.0000  0.0000
T2              -5.8111  0.0000  0.0000 11.6222  5.8111
T3              -5.8111  0.0000  0.0000  5.8111 11.6222
```

- Adjusted varaince matrix:

```
            (Intercept)      B2      B3      T2      T3
(Intercept)      6.9444 -4.1667 -4.1667 -4.1667 -4.1667
B2              -4.1667  8.3333  4.1667  0.0000  0.0000
B3              -4.1667  4.1667  8.3333  0.0000  0.0000
T2              -4.1667  0.0000  0.0000  8.3333  4.1667
T3              -4.1667  0.0000  0.0000  4.1667  8.3333
```

§3.33. HRC example: classroom questions III

The summary table of the main-effect model:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   28.000      3.112   8.997 3.71e-11 ***
B2            -3.000      3.409  -0.880 0.384122
B3             2.000      3.409   0.587 0.560730
T2             7.667      3.409   2.249 0.030097 *
T3            14.333      3.409   4.204 0.000143 ***
```

(i) Obtain the p-values for two-sided significant tests of the parameters in the main-effect model: $\alpha_{\mathrm{m2}}, \alpha_{\mathrm{m3}}, \beta_{\mathrm{m2}}, \beta_{\mathrm{m3}}$.

(ii) At level 0.05, test whether, in general, the attitude of the officers can be improved (with higher assessment scores) by increasing the training time from 10 hours to 15 hours.

Solution:

(i) The t-statistics are re-computed as follows:

$$T_{\alpha_{\mathrm{m2}}} = -3/\sqrt{8.3333} = -1.0392, \;\; T_{\alpha_{\mathrm{m3}}} = 2/\sqrt{8.3333} = 0.6928,$$
$$T_{\beta_{\mathrm{m2}}} = 7.667/\sqrt{8.3333} = 2.6558, \;\; T_{\beta_{\mathrm{m3}}} = 14.333/\sqrt{8.3333} = 4.9652.$$

§3.34. HRC example: classroom questions III (cont.)

(i) (cont.) The p-values are computed by `2*pt(-abs(T),36)`, which are given as follows:

```
             B2            B3            T2            T3
3.056274e-01 4.928678e-01 1.171402e-02 1.670833e-05
```

(ii) When the training time is increased from 10 hours to 15 hours, the general expected change of scores is $\beta_{\mathrm{m3}} - \beta_{\mathrm{m2}}$. It amounts to test

$$H_0 : \beta_{\mathrm{m3}} - \beta_{\mathrm{m2}} \leq 0 \ v.s. \ H_1 : \beta_{\mathrm{m3}} - \beta_{\mathrm{m2}} > 0.$$

The test statistic

$$\begin{aligned} T &= (\hat{\beta}_{\mathrm{m3}} - \hat{\beta}_{\mathrm{m2}})/sd_{\hat{\beta}_{\mathrm{m3}} - \hat{\beta}_{\mathrm{m2}}} \\ &= (14.333 - 7.667)/\sqrt{8.3333 + 8.3333 - 2 * 4.1667} = 2.3092. \end{aligned}$$

Note: the variances and covariance are taken from the adjusted variance matrix. The p-value is computed by `pt(-2.3092,36) = 0.01338`. H_0 is rejected at level $\alpha = 0.05$. It is concluded that the increase of training time does improve the attitude of the officers.

§3.35. DVD sales of Movies: Data and pre-treatment

- This data concerns domestic DVD sales for movies released on more than 500 screens in 2009. The data set, which is given in the file `movies2009.csv` on Canvas, contains 129 records on 10 variables including `DVD`, `Rating` and `Genre`, where `DVD` is the sales in millions of dollars, `Rating` is the rating of the DVD's with four values. `G`, `PG`, `PG-13` and `R`, `Genre` has values. `Action`, `Adventure`, `Comedy`, `documentary`, `Drama`, `Musical`, `Horror`, `Romance` and `Thriller`.
- In this analysis, we consider how the Rating and Genre of the DVDs affect the sales.
- There are only 4 records with Rating `G`, 1 record with genre `Musical` and 1 record with genre `Documentary`, there are a few missing values in variable `DVD`, these records are omitted from analysis.
- Final data set consists of 118 observations. The genres are combined into 3 categories: Action(`Action/Adventure`), Comedy (`comedy`) and Drama (`horror/romance/thriller/drama`). Rating retains 3 categories: `PG`, `PG-13` and `R`. It is a two-factor data with unbalanced cells. log DVD sales is considered as the response variable.

§3.36. DVD example: R codes for pre-treatment

```
dvd.dat0=read.csv(file="D://Rsession/movies2009.csv", header = TRUE)
dvd.dat=dvd.dat0[,c(7,9,10)] ## Extracting variables for the analysis
is.na(dvd.dat)               ## Checking for missing values
dvd.dat=na.omit(dvd.dat)     ## Deleting missing values
## Combine genre categories
  Genre = dvd.dat$Genre
  Genre[Genre=="Adventure"]="Action"
  Genre[Genre=="Horror"]="Drama"
  Genre[Genre=="Thriller"]="Drama"
  Genre[Genre=="Romance"]="Drama"
dvd.dat$Genre = Genre
## Deleting records with Rating: G, and Genres: Musical, Documentary
Rating = dvd.dat$Rating
dvd.dat=dvd.dat[Rating != "G",]
Genre = dvd.dat$Genre
dvd.dat = dvd.dat[Genre!="Musical"&Genre!="Documentary",]
## Store cleaned data
write.table(dvd.dat,file="D://Rsession/dvd1.txt", sep=" ")
```

§3.37. DVD example: fitted results

- ANOVA table of interaction model with factor order: Rating, Genre.

```
                Df Sum Sq Mean Sq F value   Pr(>F)
Rating           2  4.596  2.2979  2.5961 0.079172 .
Genre            2  8.775  4.3873  4.9566 0.008704 **
Rating:Genre     4  4.134  1.0335  1.1676 0.329176
Residuals      109 96.480  0.8851
```

- ANOVA table of interaction model with factor order: Genre, Rating.

```
                Df Sum Sq Mean Sq F value   Pr(>F)
Genre            2 12.201  6.1007  6.8924 0.001518 **
Rating           2  1.169  0.5845  0.6603 0.518741
Genre:Rating     4  4.134  1.0335  1.1676 0.329176
Residuals      109 96.480  0.8851
```

- Summary table of main-effect model:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.44981    0.20887  16.517   <2e-16 ***
RatingPG-13   -0.04356    0.23521  -0.185   0.8534
RatingR       -0.25118    0.26058  -0.964   0.3371
GenreComedy   -0.31876    0.22365  -1.425   0.1568
GenreDrama    -0.71296    0.22861  -3.119   0.0023 *
```

- Adjusted variance matrix of the main-effect model:

```
              (Intercept) RatingPG-13  RatingR GenreComedy GenreDrama
(Intercept)       0.04337    -0.03056 -0.02905    -0.02129   -0.01480
RatingPG-13      -0.03056     0.05500  0.04079    -0.00585   -0.01599
RatingR          -0.02905     0.04079  0.06750    -0.00873   -0.02120
GenreComedy      -0.02129    -0.00585 -0.00873     0.04972    0.02809
GenreDrama       -0.01480    -0.01599 -0.02120     0.02809    0.05195
```

§3.38. DVD example: classroom questions

(i) Test whether there is significant interaction effect between Genre and Rating and whether Genre and Rating have significant main effects. Draw conclusions based on the p-values of the tests.

(ii) Test whether among the three genres, Action, Comedy and Drama, any two have a significant difference in their sales.

(iii) Test whether Actions have significantly better sales than Dramas.

Solution:

(i) The F-statistic value an the p-value for the interaction effect can be obtained form either of the ANOVA tables of the interaction model. $F = 1.1676,\ p = 0.329176$. There is no strong evidence for the significance of the interaction effect. The interaction effect is not significant.

For the main-effect of Genre, the first ANOVA table is used. The p-value of the test is 0.008704. Genre has a significant main-effect since the small p-value is a strong evidence for the significance.

For the main-effect of Rating, the second ANOVA table is used. The p-value of the test is 0.518741. There is no strong evidence to support the significance, the main-effect of Rating is not significant.

§3.39. DVD example: classroom questions (cont.)

(ii) Let β_2 and β_3 denote the parameters for Comedy and Drama respectively. It amounts to test whether β_2, β_3 and $\beta_2 - \beta_3$ are sigficantly nonzero. The test statistics are computed using the estimates in the summary table of the main-effect model and the variances and covariances in the adjusted variance matrix.

$$T_1 = -0.31876/\sqrt{0.04972} = -1.4295,$$
$$T_2 = -0.71296/\sqrt{0.05196} = -3.1277,$$
$$T_3 = \frac{-0.71296 + 0.31876}{\sqrt{0.04972 + 0.05196 - 2 * 0.02809}} = -1.8480.$$

The p-values are computed by `T=c(T1,T2,T3); pt(-abs(T),109)` as $0.0779, 0.0011, 0.0337$. All the pairs have significant differences, though the evidence for the difference between Comedy and Action is slightly weaker.

(iii) It amounts to test $H_0 : \beta_3 \geq 0$ *v.s.* $H_1 : \beta_3 < 0$. The value of the test statistic is given by $T_2 = -3.1277$, the p-value $P(t \leq -3.1277)$ is $0.0011/2 = 0.00055$. Actions does have better sales than Dramas.